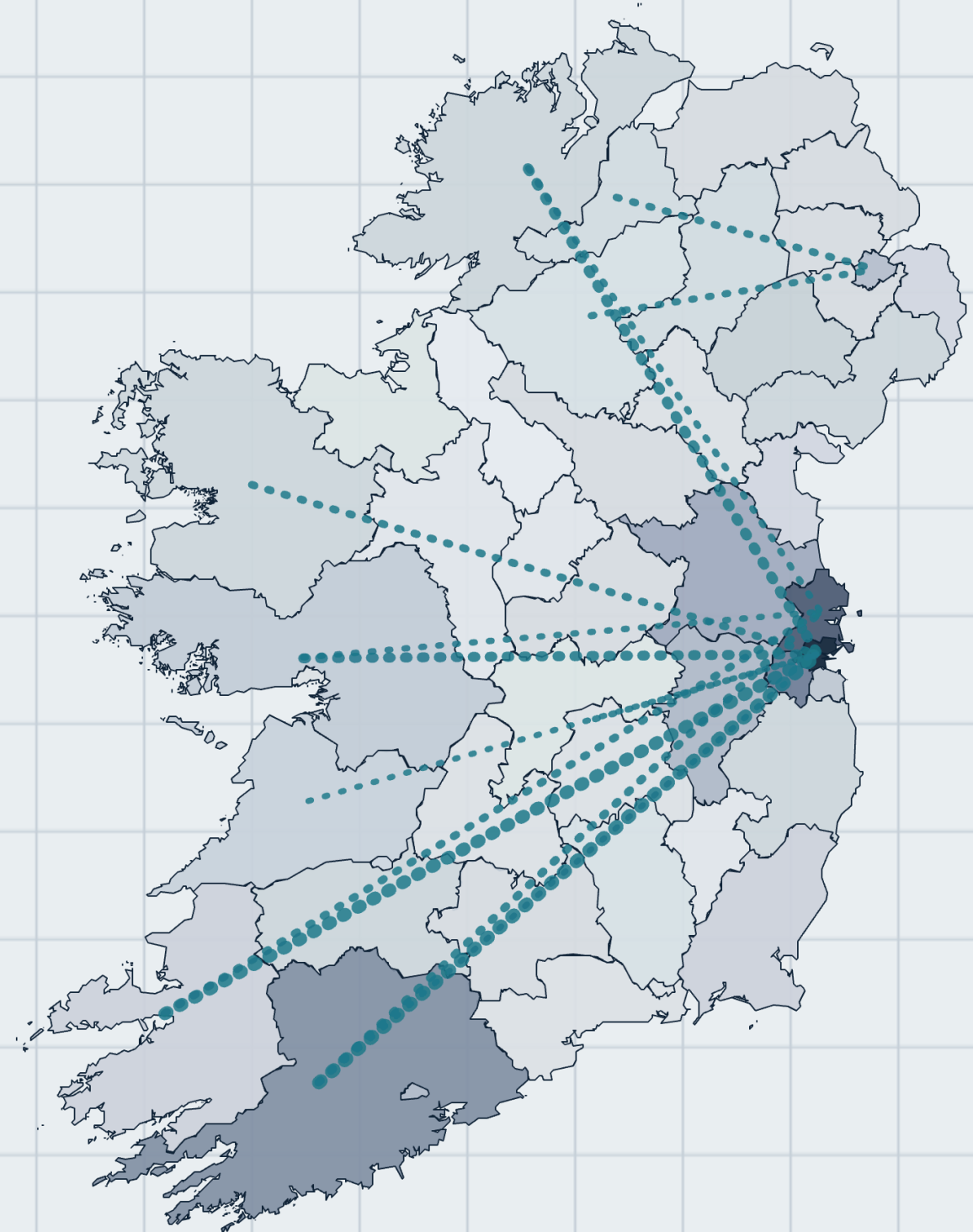


HACKATHON 2026 • EIRGRID + CSO OPEN DATA

---

# GridSense

Forecast where Ireland's load will peak.  
Route renewables there first.



SECTION 01 / 07

# The problem

Wind gets switched off while the grid says no.

---

The problem

What GridSense does

Who it's for

Impact

Live demo

What's next

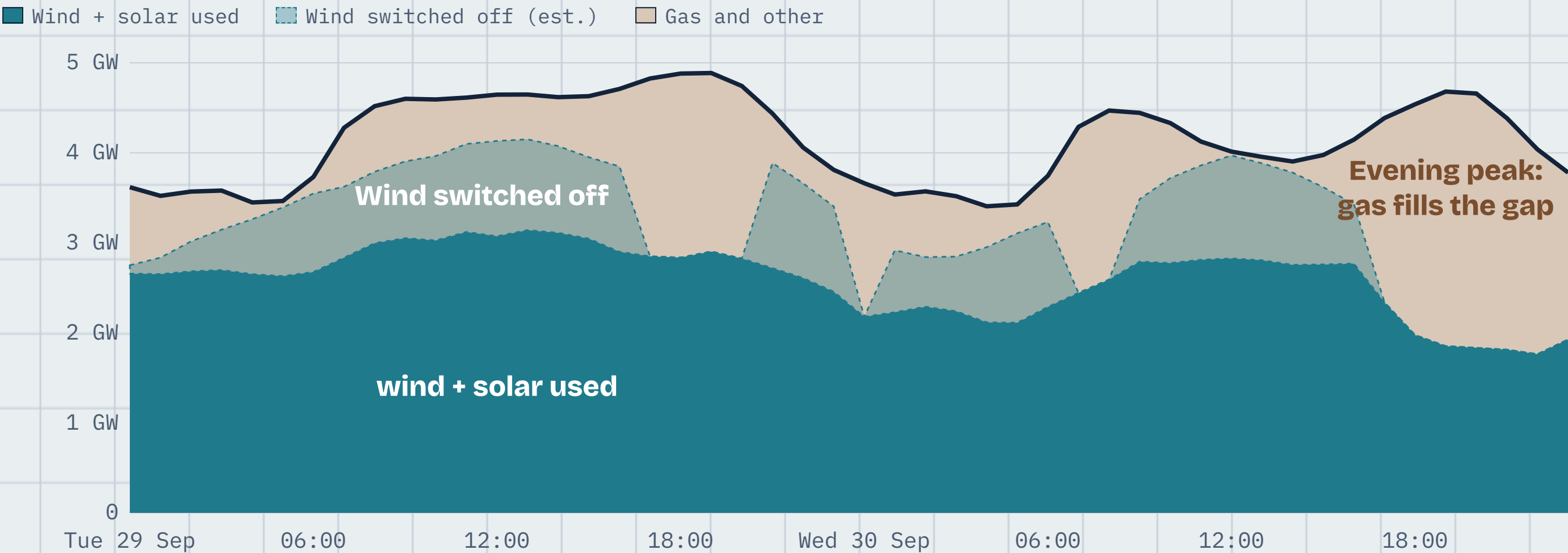
Technical details

THE PROBLEM

# Wind gets switched off. Then gas runs the peak.

37.6%

island wind + solar share, 12 months

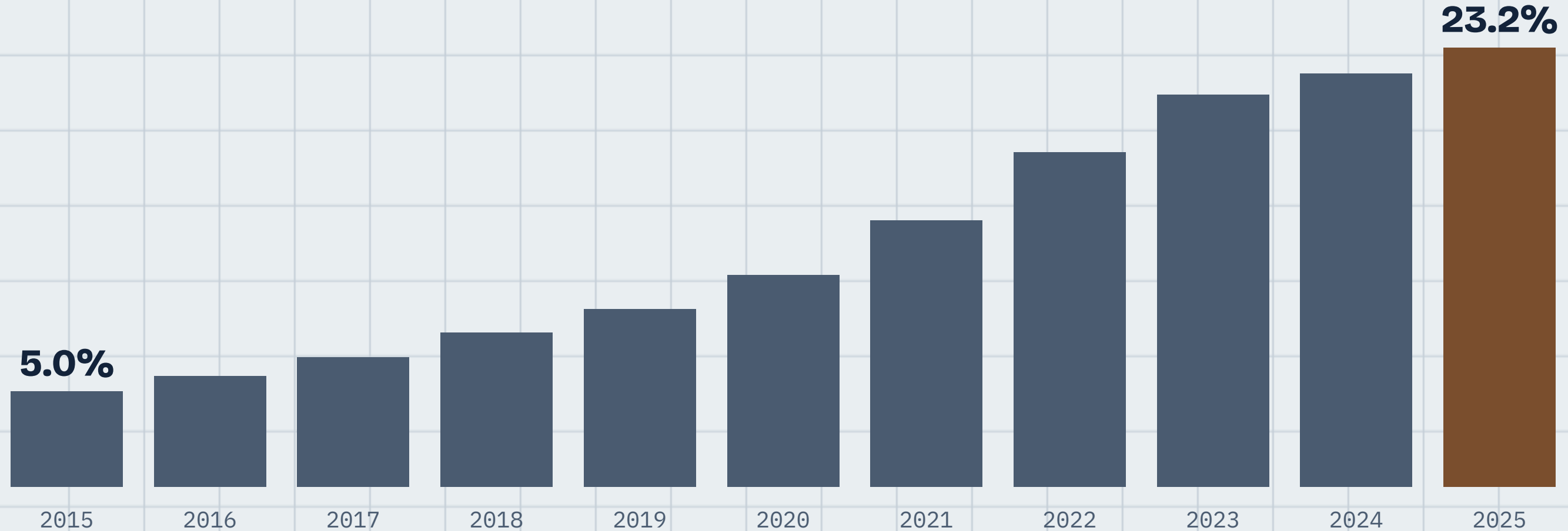


EirGrid Smart Grid Dashboard, ROI hourly, 29-30 Sep 2026. Switched-off wind estimated as wind forecast - actual when the grid is  $\geq 60\%$  renewable.

## THE PROBLEM

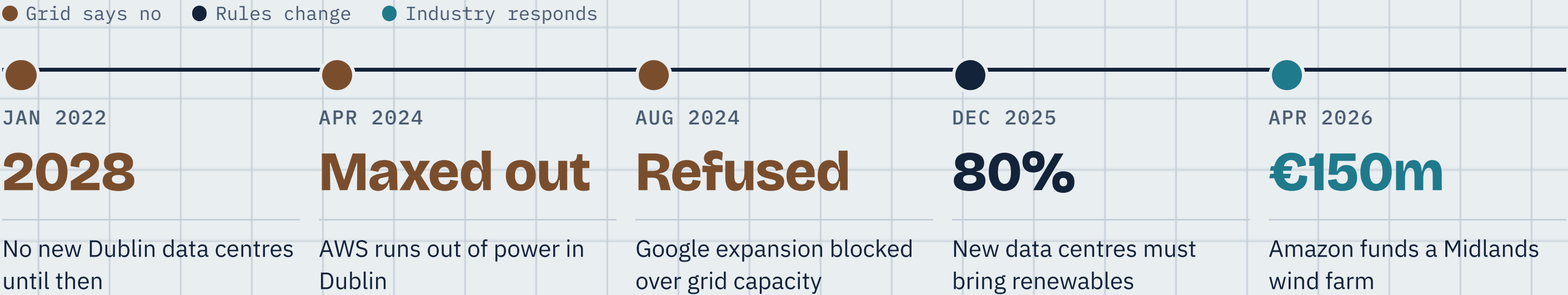
Data centres now use **23.2%** of the Republic's electricity.

**6.2×**  
more than in 2015



CSO MEC02: data-centre share of ROI metered electricity, by year, 2015–2025.

# The grid said no. The rules changed.



**Where GridSense fits:** hour-by-hour, per-area proof of renewable use.

Sources: RTÉ / DCD, Jan 2022 • The Register via DCD, 10 Apr 2024 • Techerati, Sep 2026 • CRU decision via William Fry, 15 Dec 2025 • Irish Examiner, 27 Apr 2026

SECTION 02 / 07

# What GridSense does

Forecast the peak. Route wind to it first.

---

The problem

What GridSense does

Who it's for

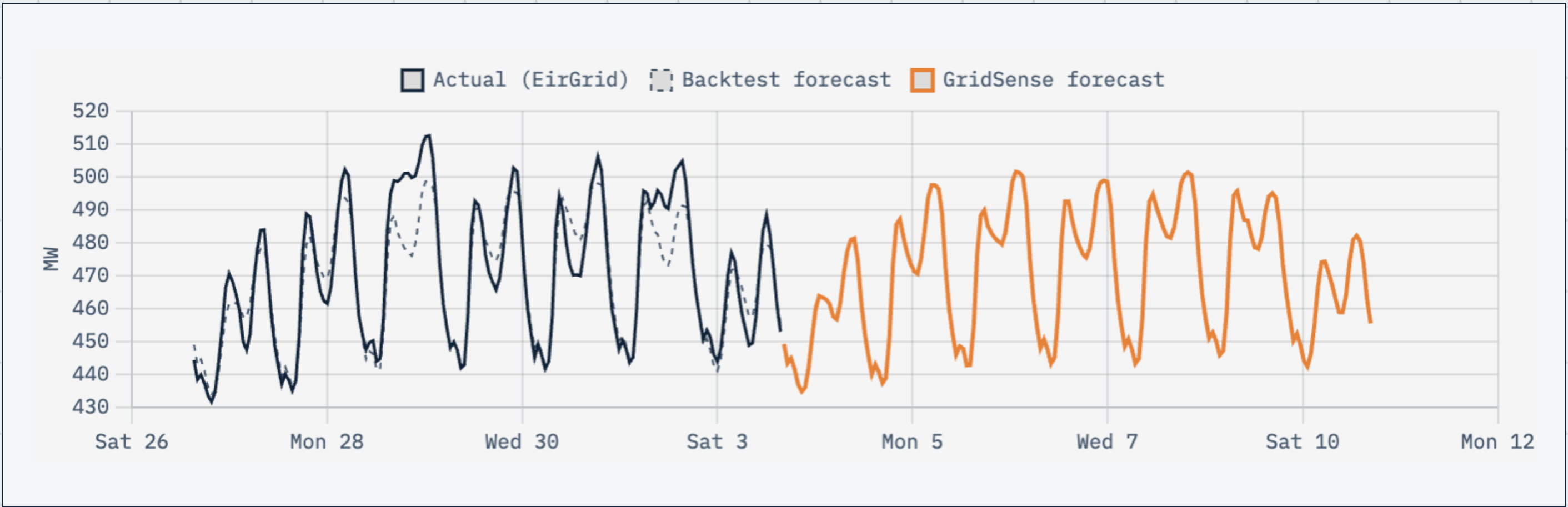
Impact

Live demo

What's next

Technical details

# We forecast every area's peak, a week ahead.



**2.8%**

backtest error, ROI

**45**

areas, hourly, 7 days

**696 MW**

Dublin City, Tue 18:00



# Then we book spare wind for the biggest peak.

Cork County → Dublin City

485 MWh

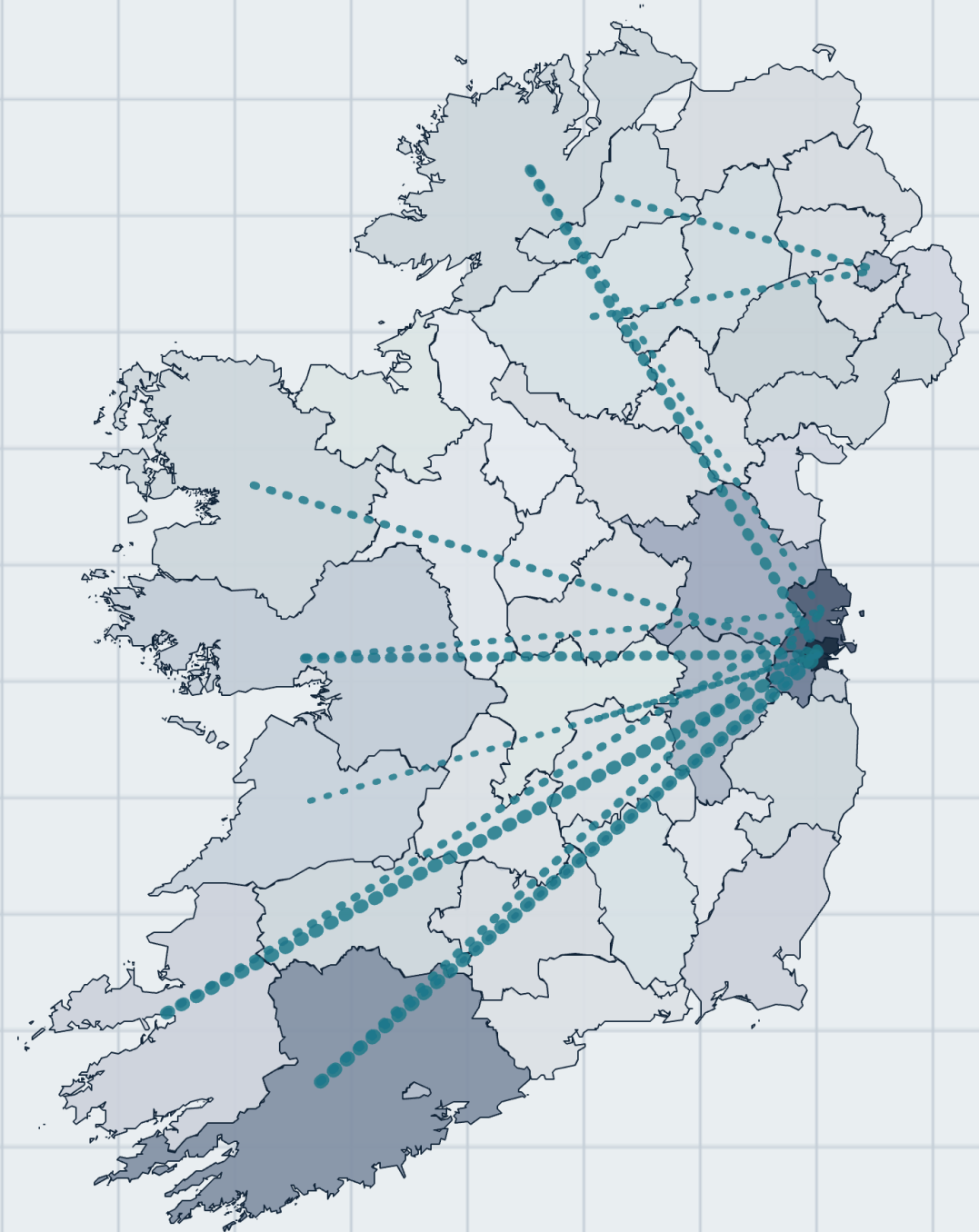
Kerry → Dublin City

461 MWh

Donegal → Dublin City

364 MWh

**12.8 GWh** planned this week





# And we prove the cut, area by area.

## -9.3 kt CO<sub>2</sub>

AVOIDED WITH GRIDSENSE • 12 MONTHS

~~670.3 kt~~ → **661.0 kt** **-1.4%**

Fingal gets **82.3 GWh** more wind, delivered in its forecast peak hours, when the grid would otherwise burn gas. Wind + solar share goes from **37.6%** to **39.6%**. The load stays the same and nothing changes on site.

Fingal, Ireland's data-centre hub • 12 months • live dashboard

SECTION 03 / 07

# Who it's for

Data centres, government, new industry,  
households.

---

The problem

What GridSense does

Who it's for

Impact

Live demo

What's next

Technical details

# Same workload. More wind behind it.

---

**1,845 MW**

forecast peak, this week

---

**+299 GWh**

recovered wind at peak hours

---

**-33.8 kt**

CO<sub>2</sub> cut, from 2.36 Mt

# What a data centre gets.

**24/7**

Hourly-matched carbon-free  
evidence

**Scope 2**

Location-based figures, per site

**80%**

An answer to the CRU rule

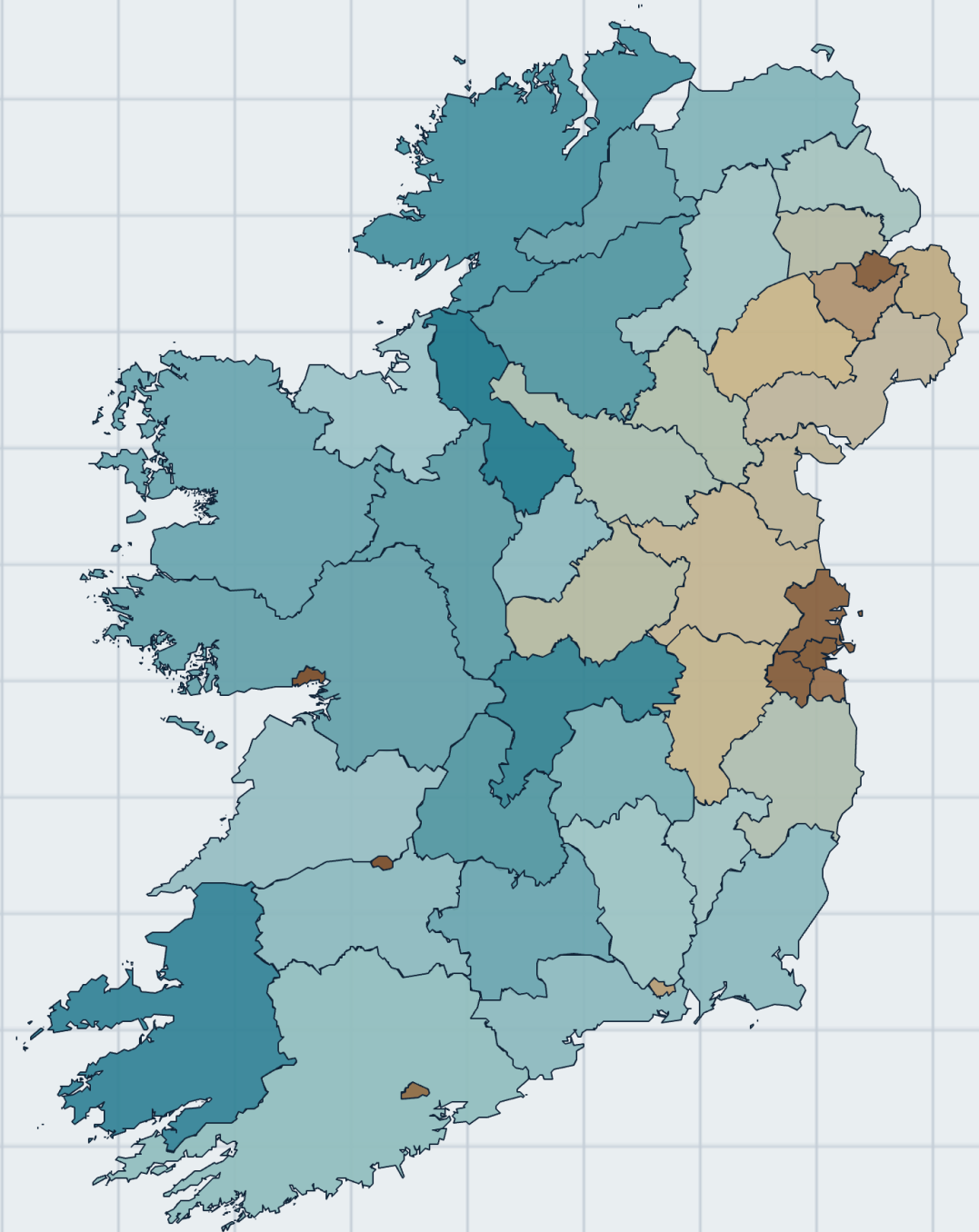
No change to workloads or SLAs.

# The west makes the wind. Dublin uses it.

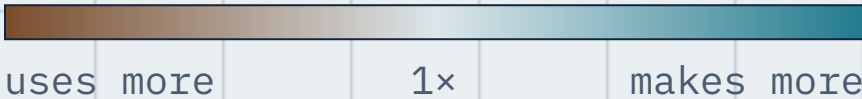
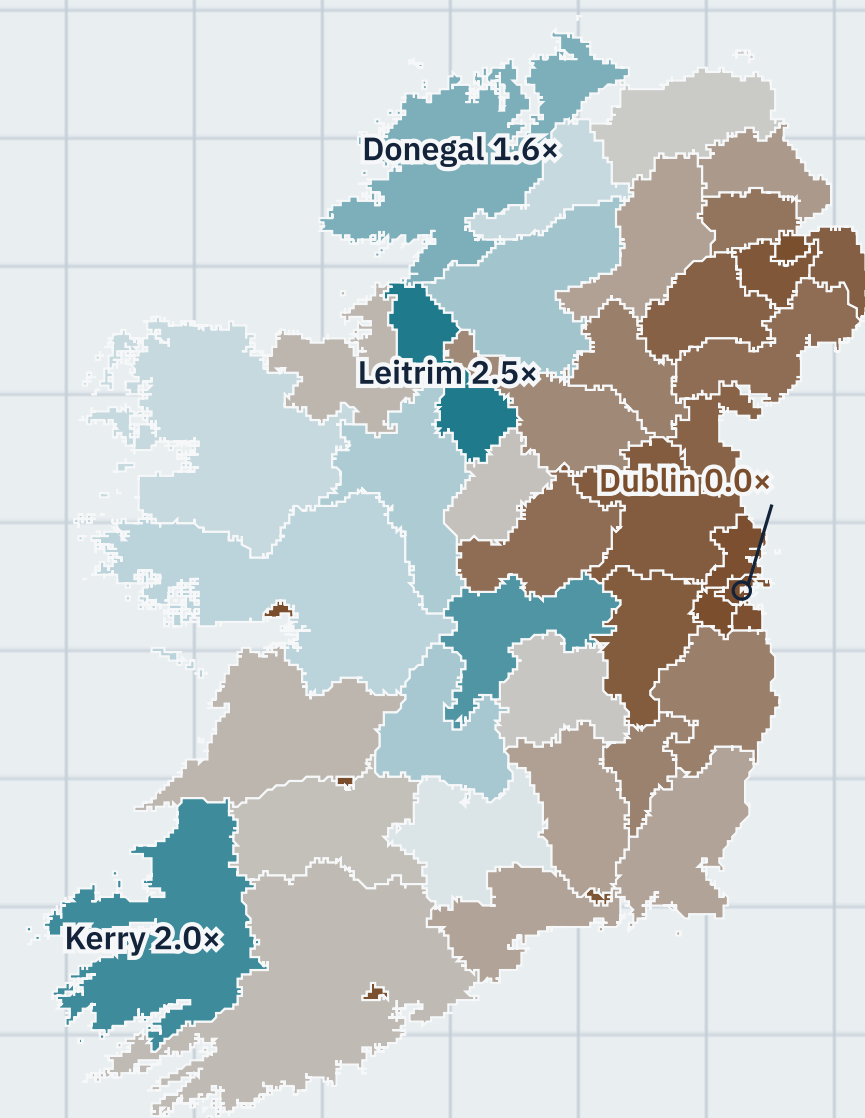
 **Leitrim 1.9× • Kerry 1.6×**  
generate more than they use

 **Dublin ~0.01×**  
imports almost all of it

Local generation ÷ use, last 3 months. Where the next  
wind farm and cable should go.



# Build where the wind is.



2.5x

Leitrim makes 2.5x the renewables it uses

0.09x

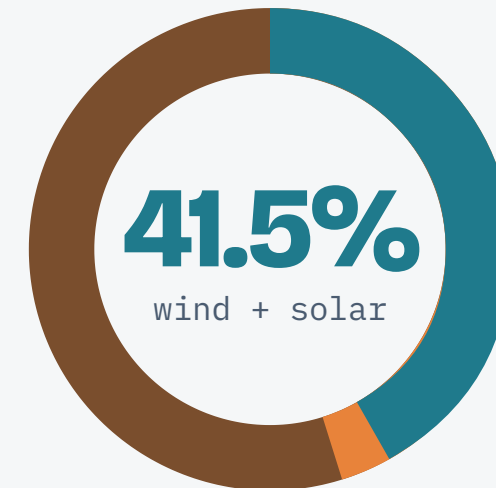
Meath and Kildare: the data-centre corridor



Site near the surplus, not the congestion

"How much of my  
power was  
renewable?"

DUBLIN CITY • LAST MONTH



**44.8%** with GridSense

**120 kg** CO<sub>2</sub> per resident



SECTION 04 / 07

# Impact

Twelve months of real EirGrid data.

---

The problem

What GridSense does

Who it's for

Impact

Live demo

What's next

Technical details

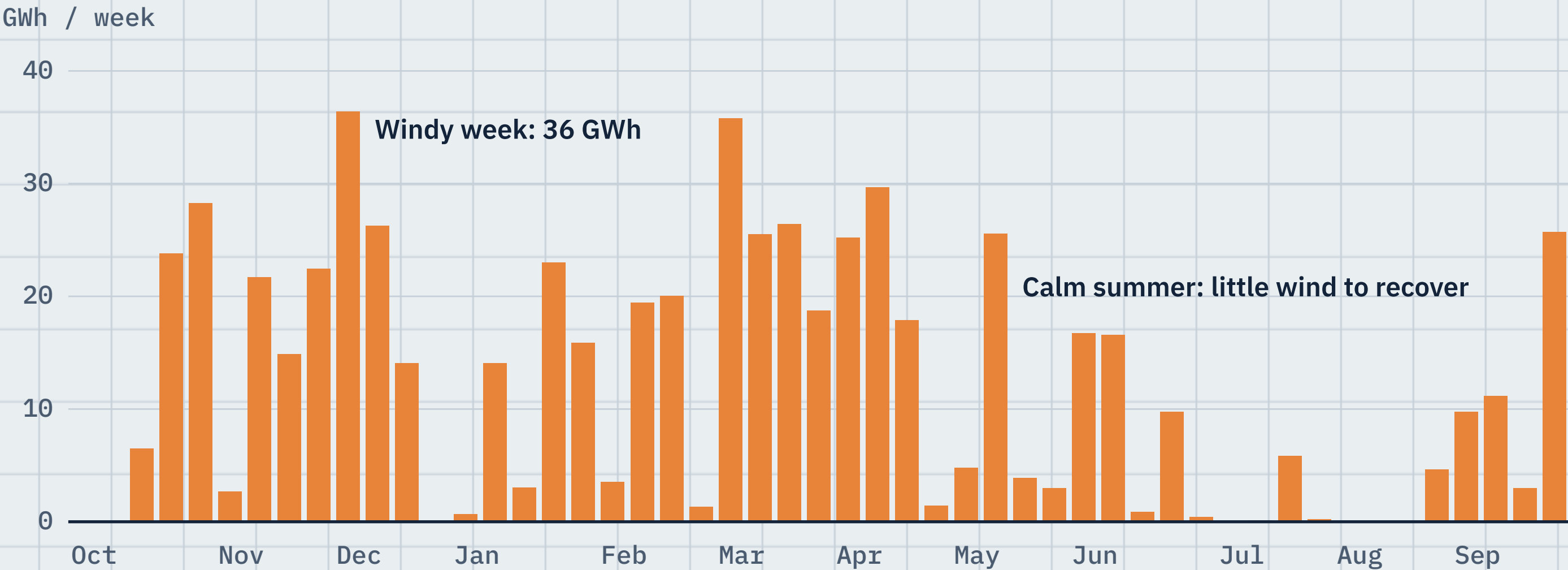


# What a year of GridSense does.



Modelled on real EirGrid data, Oct 2025 – Oct 2026. Same load, same weather.

# Windy weeks do the heavy lifting.



SECTION 05 / 07

# Live demo

The real dashboard, all 45 areas.

► Watch the demo video · [youtube.com/watch?v=chCNufY8-1Y](https://youtube.com/watch?v=chCNufY8-1Y)

---

The problem

What GridSense does

Who it's for

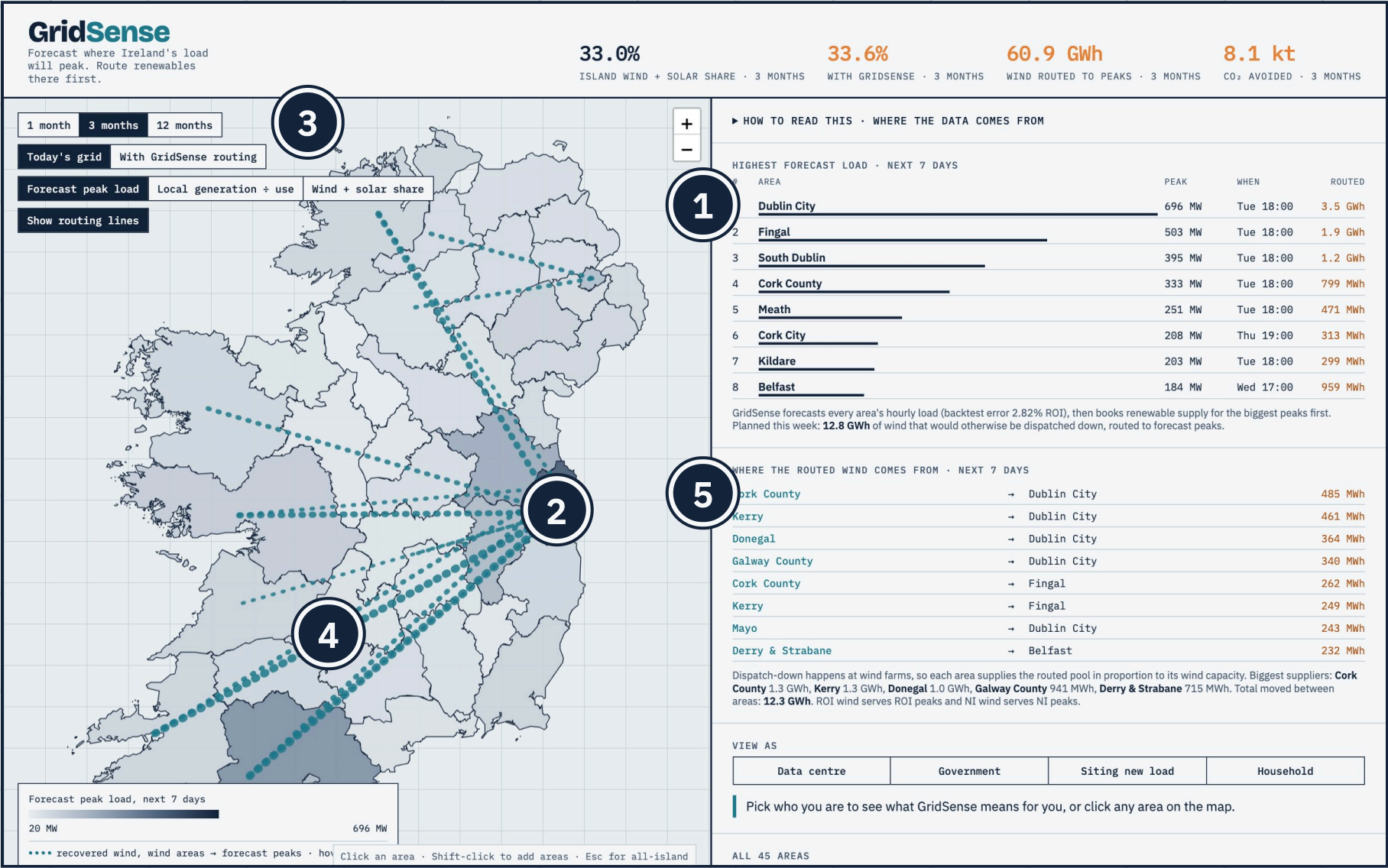
Impact

Live demo

What's next

Technical details

# Follow the wind.



SECTION 06 / 07

# What's next

From prototype to live routing.

---

The problem

What GridSense does

Who it's for

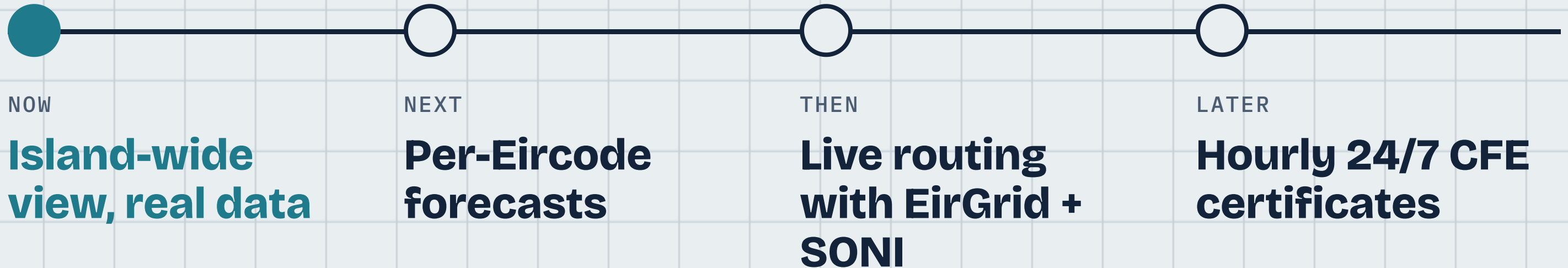
Impact

Live demo

What's next

Technical details

# From open data to live routing.



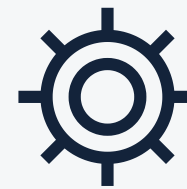
Add-on: flex signals for customers who want more

# Three things get us live.



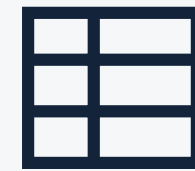
## Node-level wind data

EirGrid + SONI



## A data-centre pilot

Fingal or Meath



## A policy sponsor

forecast-led grid scheduling



# Green-hour tokens.



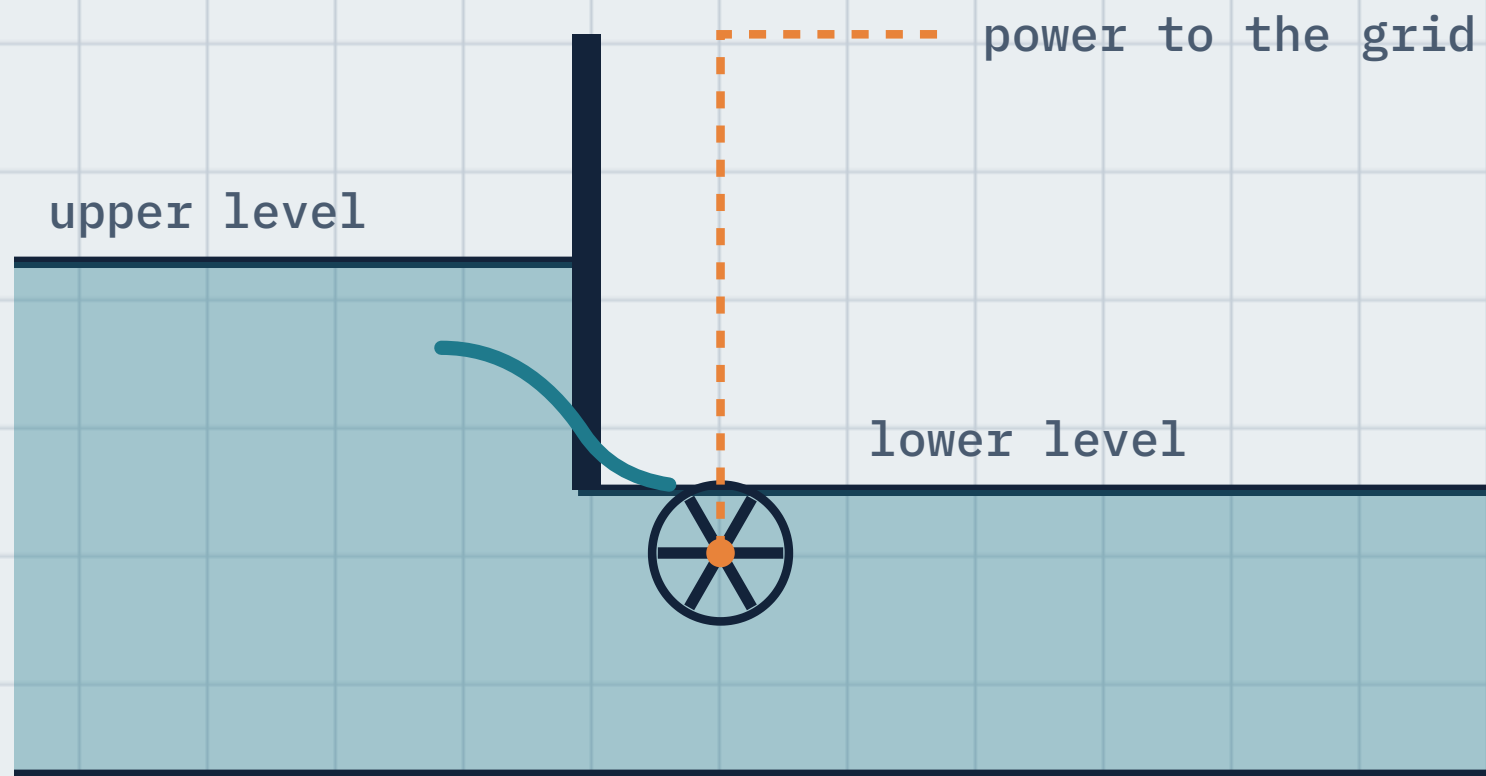
— Fund renewables, earn priority green hours

— A cap per company: no cornering the wind

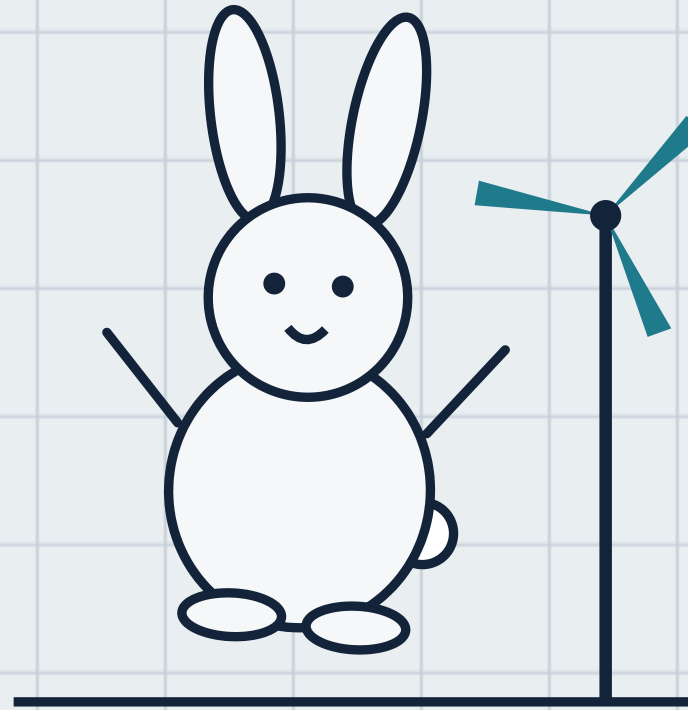
— Every token is an auditable hourly record



# Micro-turbines on Ireland's canals.



- Turbines at locks and weirs, where water drops
- Steady output, day and night
- GridSense picks the sites



# Thank you.

This bunny's moves are powered by wind  
that would otherwise have been switched off.

Questions?

## APPENDIX

# Technical details

The data, the forecast, the routing. Read at your own pace.

---

The problem

What GridSense does

Who it's for

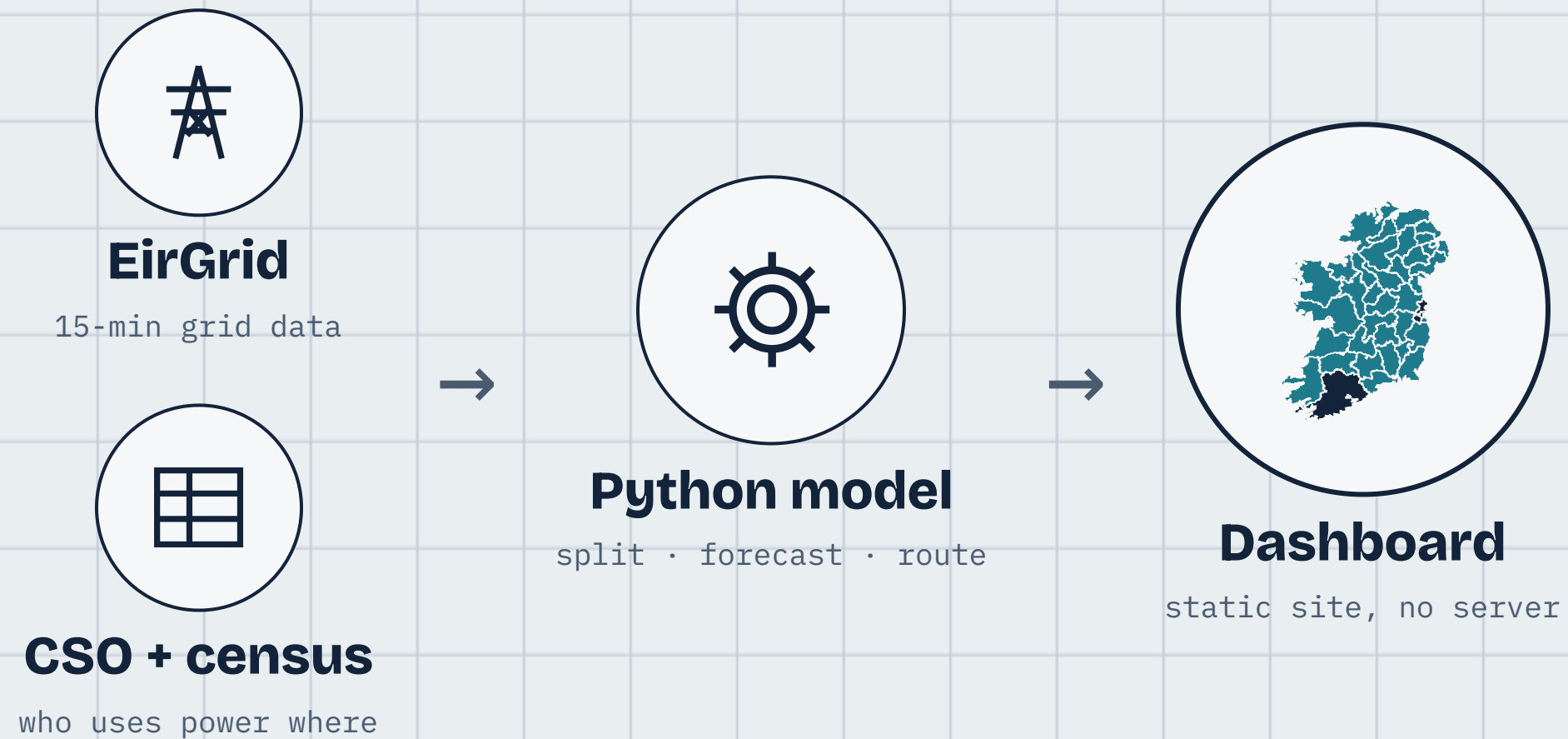
Impact

Live demo

What's next

Technical details

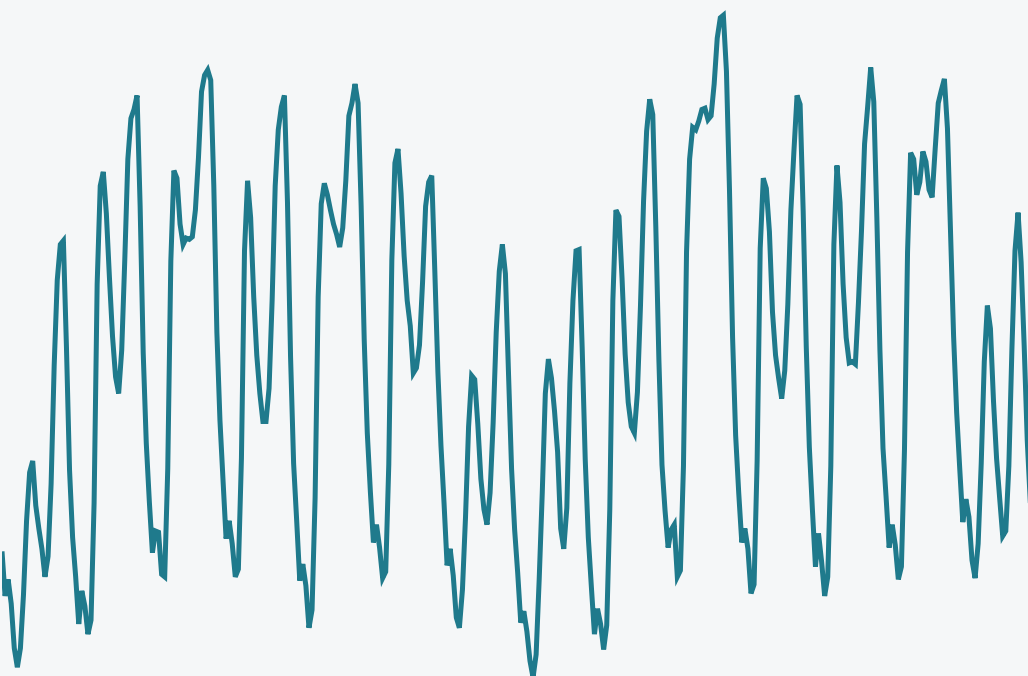
# Open data in. One dashboard out.



# All public. All real.

## EirGrid

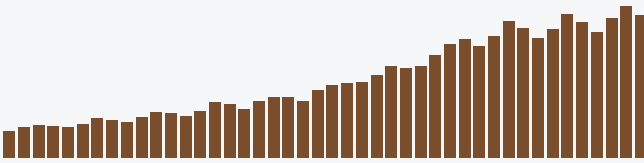
demand · wind · solar · CO<sub>2</sub> · 15-min



ROI demand, last 14 days

## CSO MEC02

data centres: 23.2% of ROI power



2015Q1 → 2025Q4

## NISRA

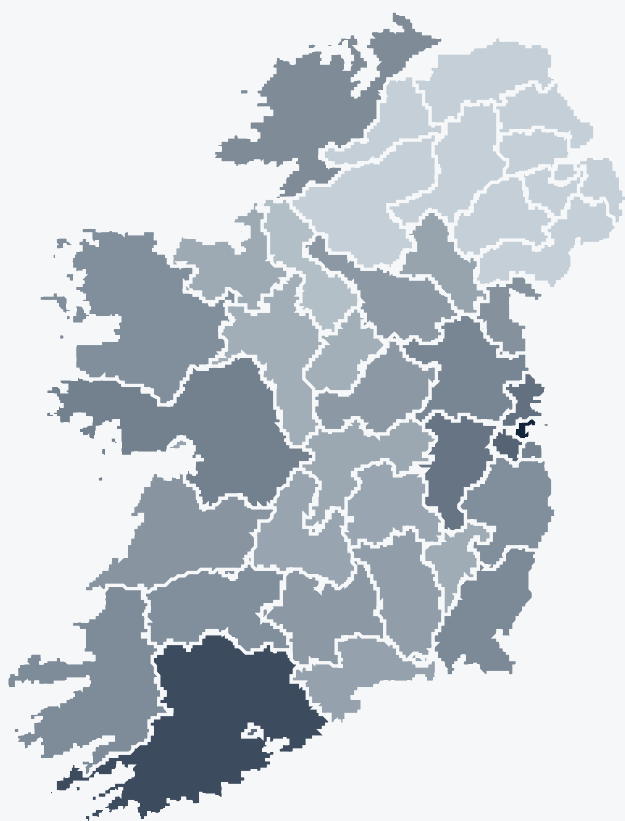
NI population

## OSi · OSNI

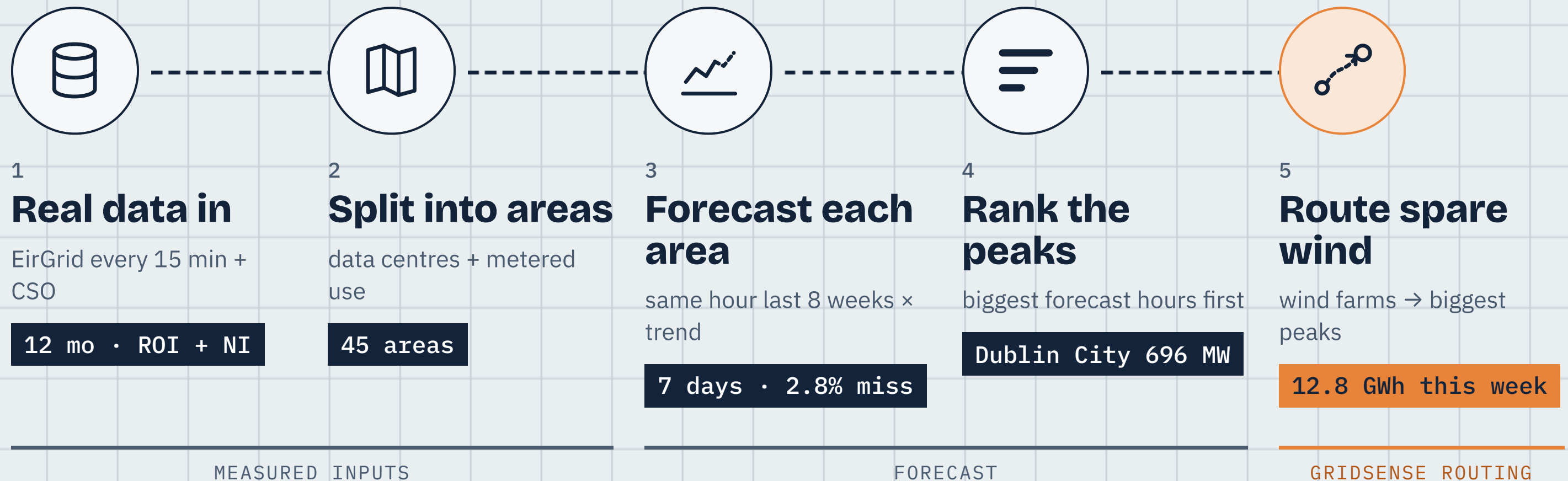
boundaries

## CSO MEC03

use by county

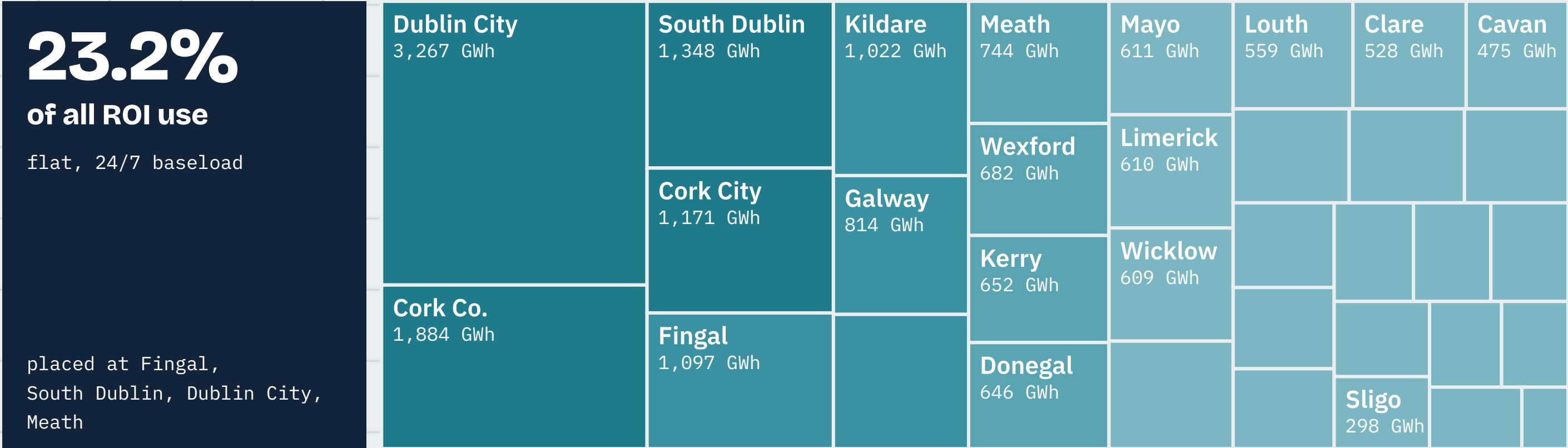


# Five steps from open data to a routing plan.



# Step 2: share the national load out to 45 areas.

CSO MEC02 · DATA CENTRES    CSO MEC03 · METERED USE PER COUNTY + DUBLIN POSTCODE, 2025



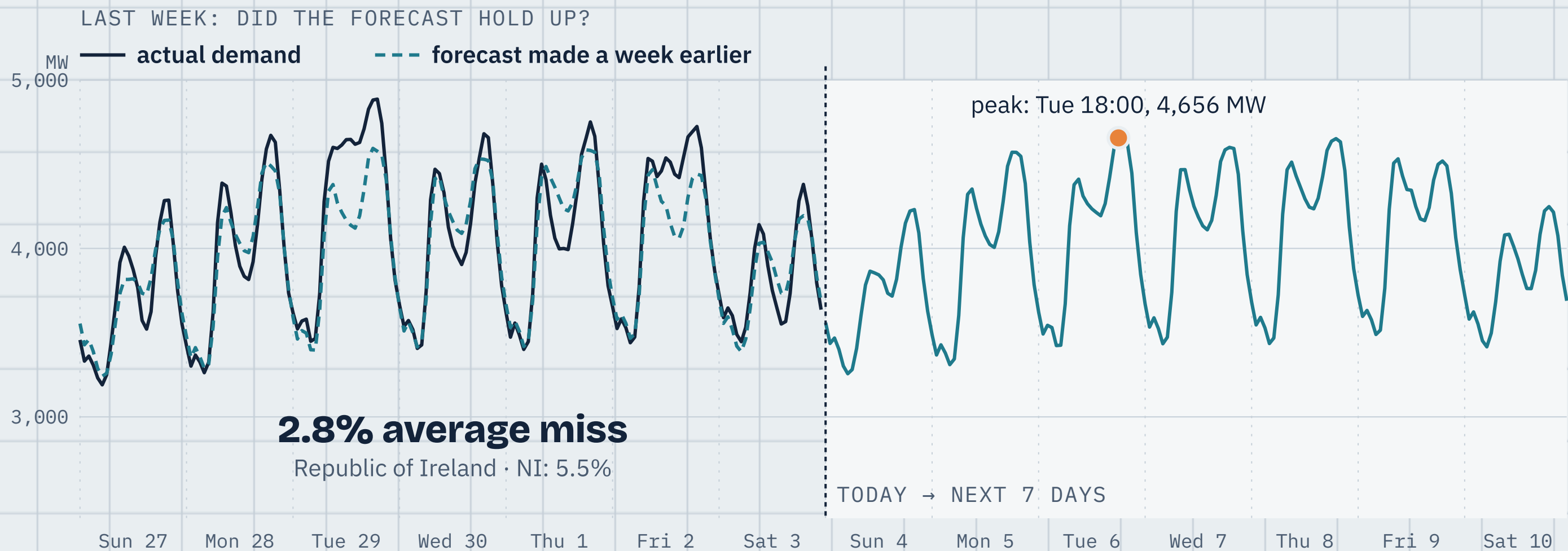
tile size = electricity really used

+ NI: 11 districts, by population

✓ adds up to EirGrid demand, ±1%



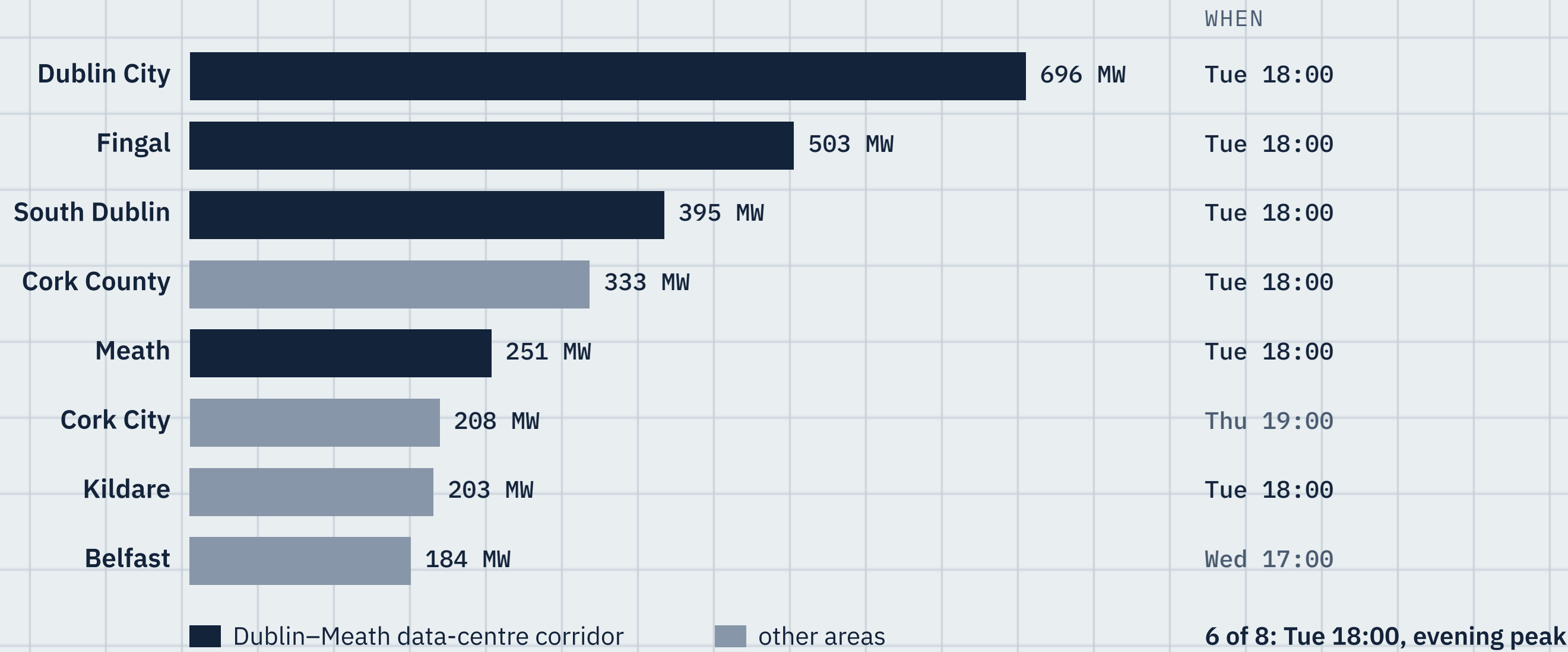
# Step 3: forecast every hour of next week.



how: average of the same hour over the last 8 weeks × last week's trend

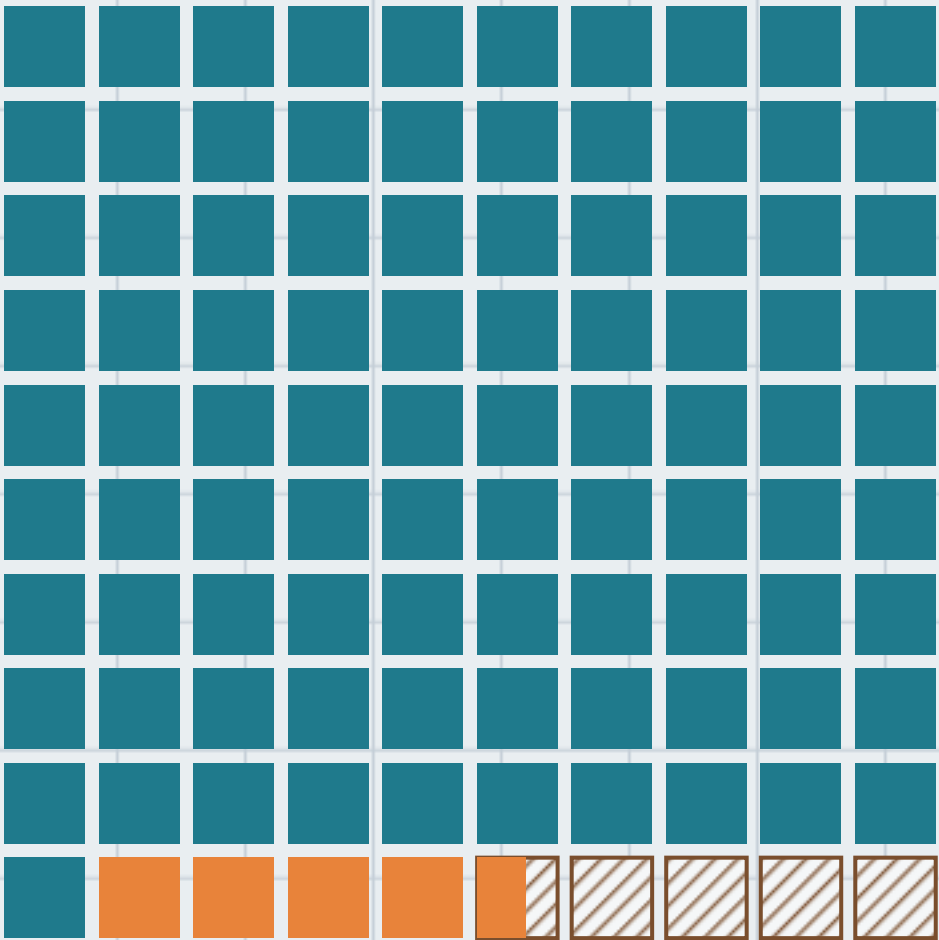


# Step 4: rank where next week's peaks hit.



The biggest peaks get first call on spare wind →

# Step 5a: find the wind that gets switched off.



every square = 1% of ROI wind, 12 months

**91% reaches customers**

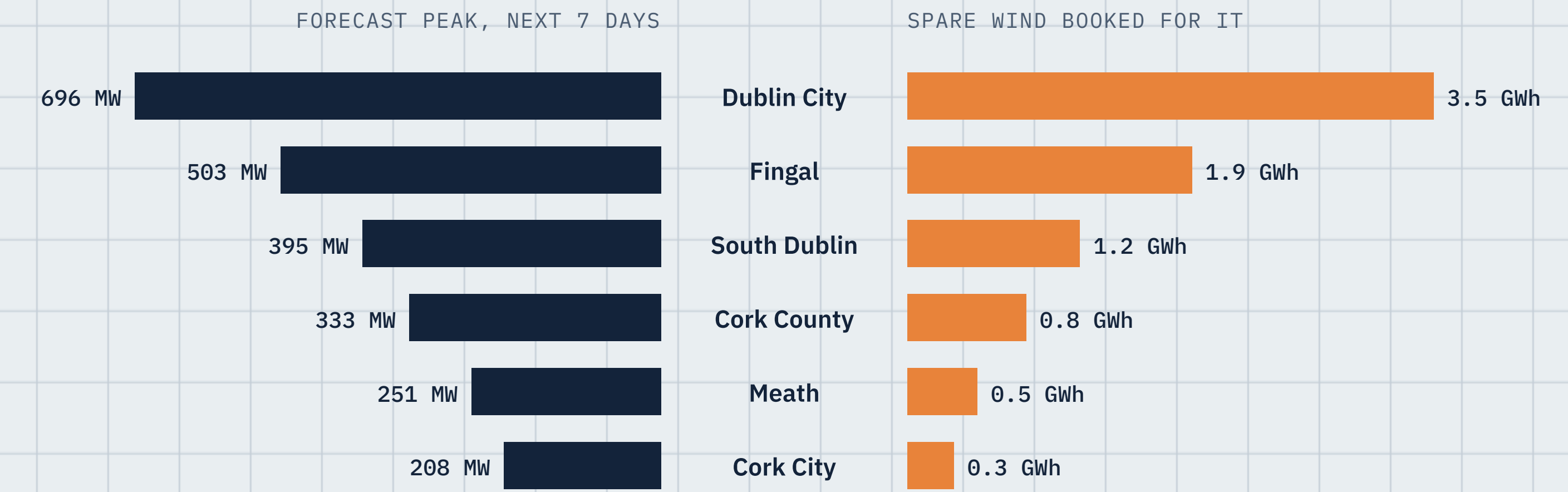
wind that the grid could absorb

**9.2% dispatched down**

wind farms told to stop: forecast – actual, when the grid is  $\geq 60\%$  renewable  
EirGrid reports  $\sim 10\%$  ✓

Half of it ( $\approx 4.6\%$ ) is grid congestion.  
GridSense schedules that part into forecast peaks.

# Step 5b: share it out, biggest peaks first.



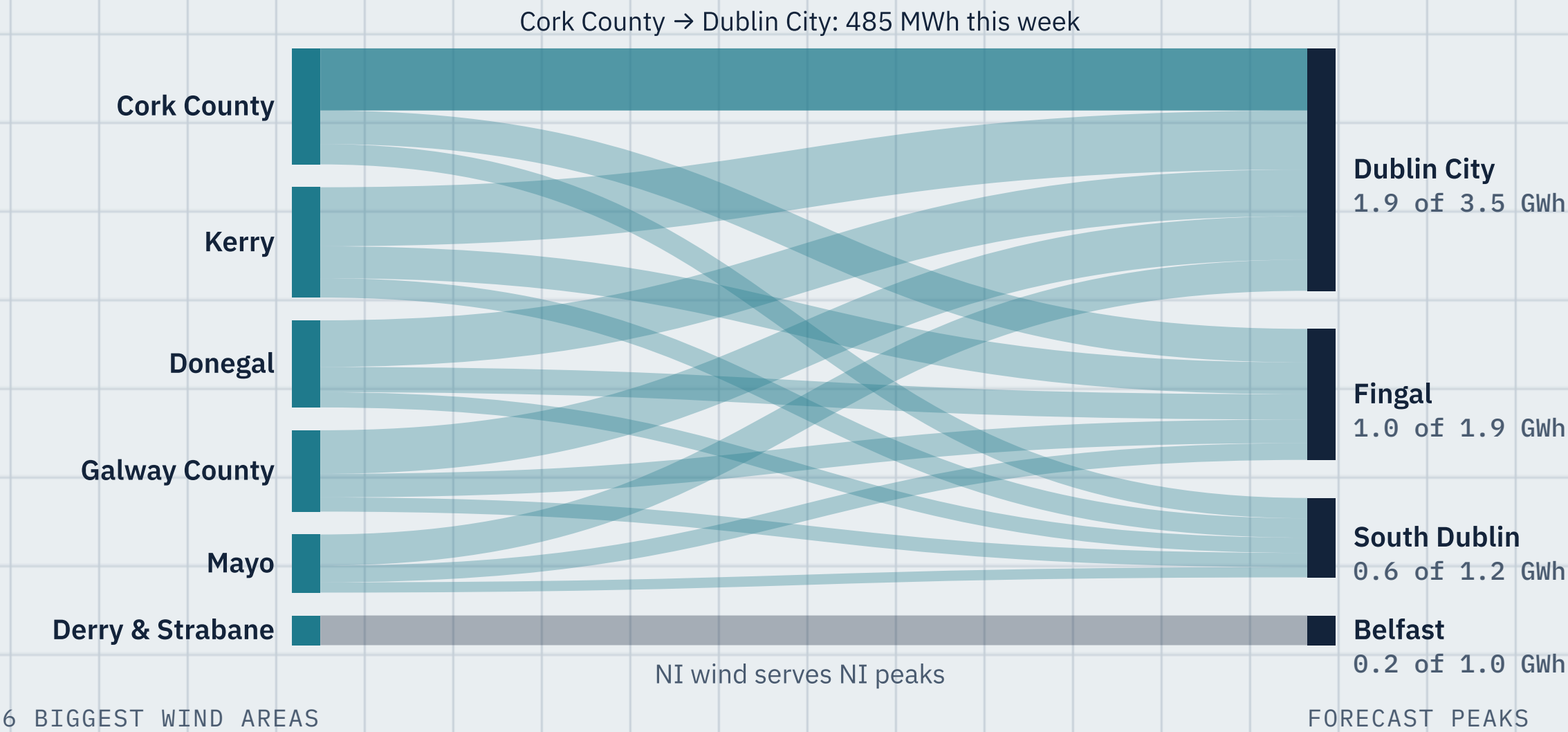
Share grows with peak<sup>2</sup>: Dublin City's peak is 1.4× Fingal's, so it gets 1.8× the wind. Everyone gets some.

into each day's 6 highest forecast hours

capped at the area's gas-fired load in those hours

leftover re-shared

# Step 5c: every flow has a source and a number.

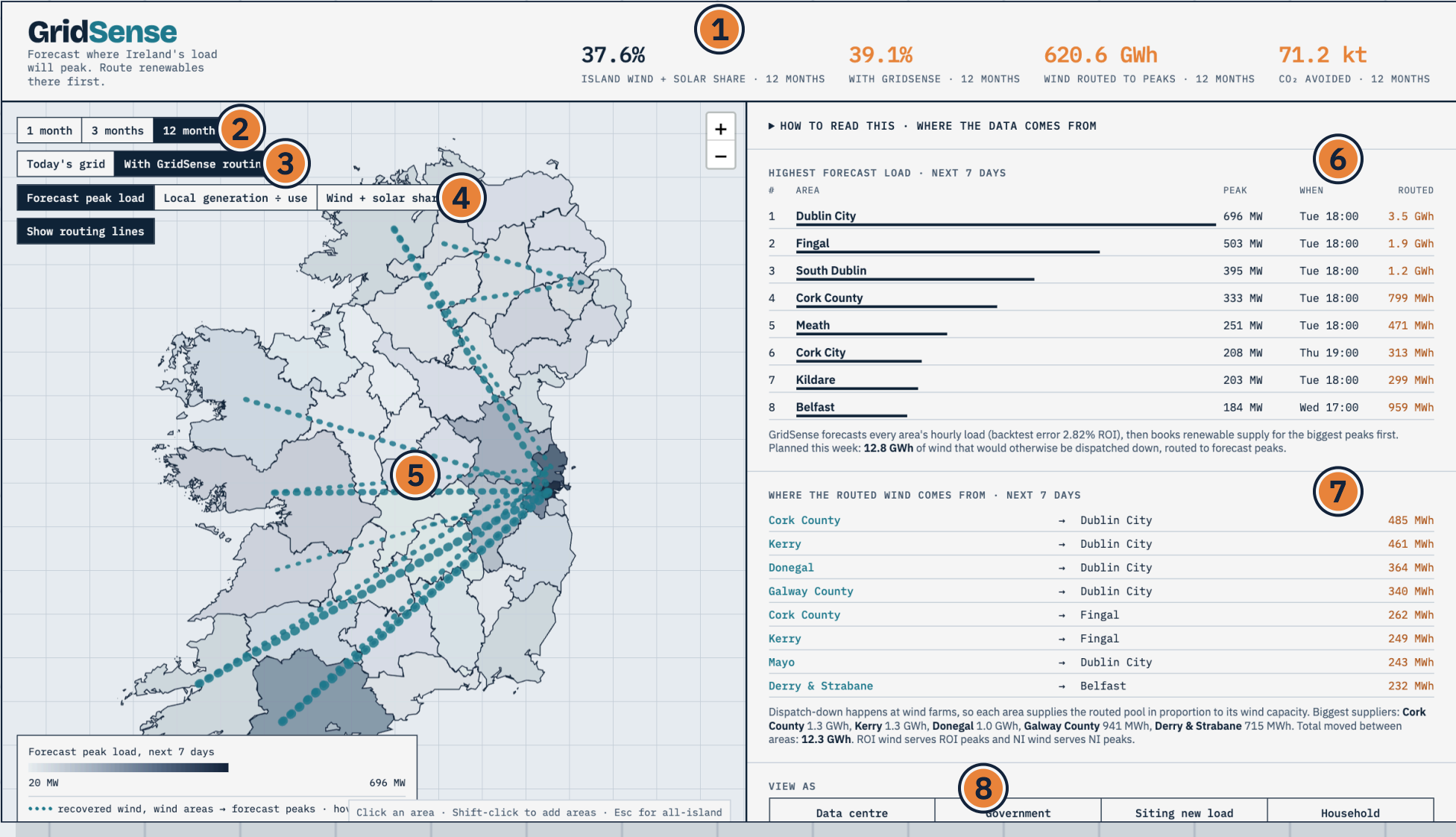


thicker = more energy, next 7 days

each wind area supplies in line with its wind capacity

12.8 GWh planned

# One screen: where peaks hit, where wind comes from.



# Click any area: see what GridSense saves there.

SELECTED AREA

Fingal

2025-10-02 → 2026-10-03 · Republic of Ireland

37.6% → 39.6% ①

wind + solar share today → with GridSense routing

-9.3 kt CO<sub>2</sub> ②

AVOIDED WITH GRIDSENSE · 12 MONTHS

~~670.3 kt~~ → 661.0 kt -1.4% ③

Fingal gets 82.3 GWh more wind, delivered in its forecast peak hours, when the grid would otherwise burn gas. Wind + solar share goes from 37.6% to 39.6%. The load stays the same and nothing changes on site.

4.11 TWh

electricity used

670.3 kt

CO<sub>2</sub> footprint today

503 MW

forecast peak, next 7 days

④

- ① Wind + solar share: today → with GridSense
- ② CO<sub>2</sub> avoided, this period
- ③ Footprint before → after
- ④ Next week's forecast peak